

CS07340 DESIGN AND ANALYSIS OF ALGORITHMS

WEEK 2 QUIZ (8 POINTS)

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1) (4 points) The following are True/False questions about algorithm complexity. Clearly mark either $\textcircled{T}$ (for True) or $\textcircled{F}$ (for False).

- ☒ $\textcircled{F}$ The *Traveling Salesman Problem* is NP hard, that is, no polynomial algorithm in the input size n is known.
- ☒ $\textcircled{F}$ The *Knapsack Problem* is NP hard, that is, no polynomial algorithm in the input size n is known.
- $\textcircled{T}$ ☒ $\textcircled{F}$ The *Assignment Problem* is NP hard, that is, no polynomial algorithm in the input size n is known.
- $\textcircled{T}$ ☒ $\textcircled{F}$ Evaluating a polynomial $p(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$ for a given number x_0 is NP hard, that is, no polynomial algorithm in the input size n is known.

2) (4 points) The following are questions about programming paradigms.

a) What is the purpose of *Exhaustive Search*?

The purpose of Exhaustive Search is to search for the key by comparing it to every element in the array.

b) What is the purpose of *Depth-First Search*?

The purpose of Depth-First Search is to divide the array into sections and search sections.

c) What is the purpose of *Breadth-First Search*?

The purpose of Breadth-First search is to search at first instance of finding key.

d) *Depth-First Search* and *Breadth-First Search* need a ADT or data structure as input. What type of input is required for these two traversal algorithms?

An array is required for both of these traversal algorithms.